

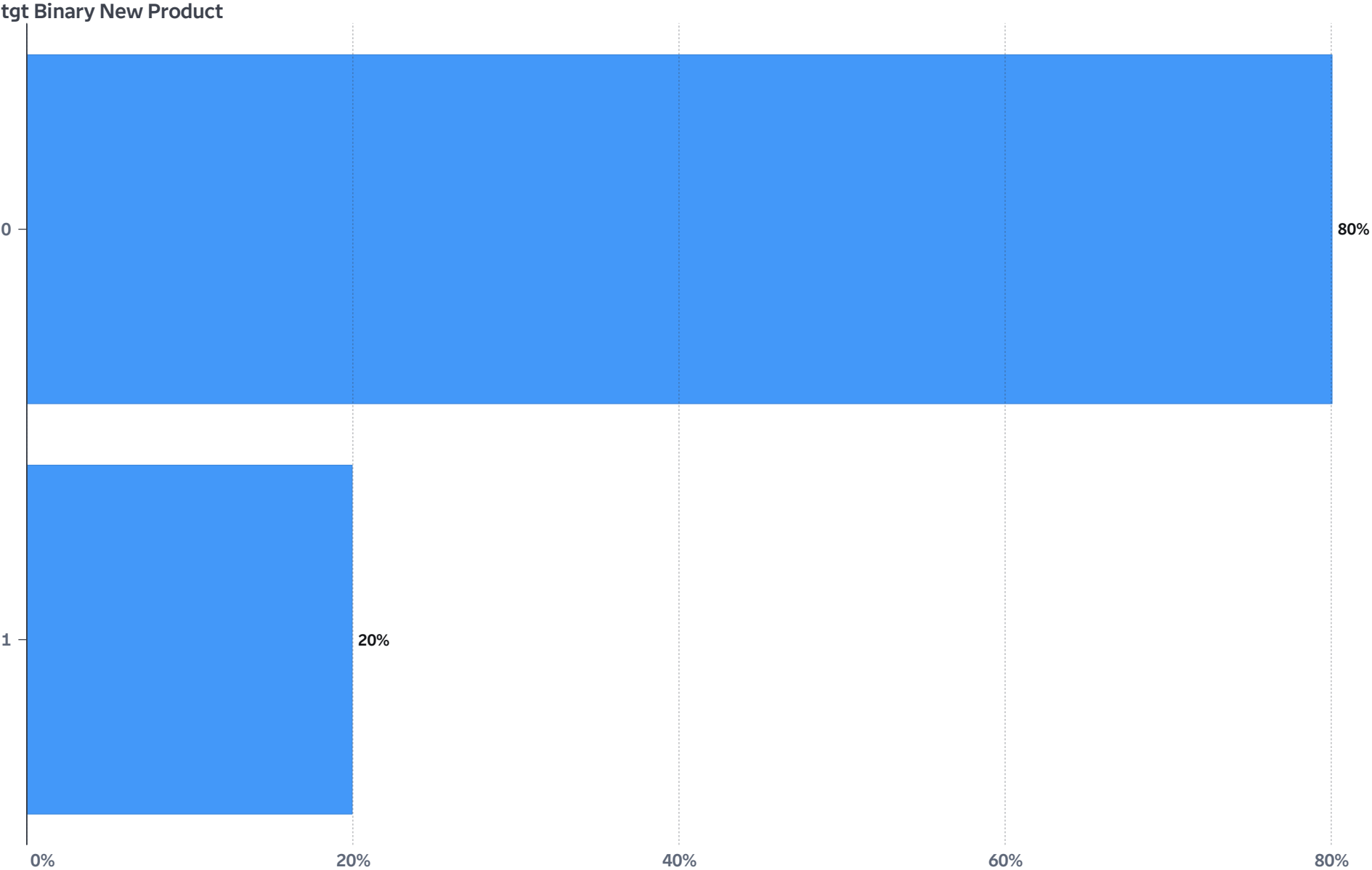
Bank Analysis

Creation Date: Monday, September 21, 2026, 05:48:02 PM

Author: 24bba70187@cuchd.in

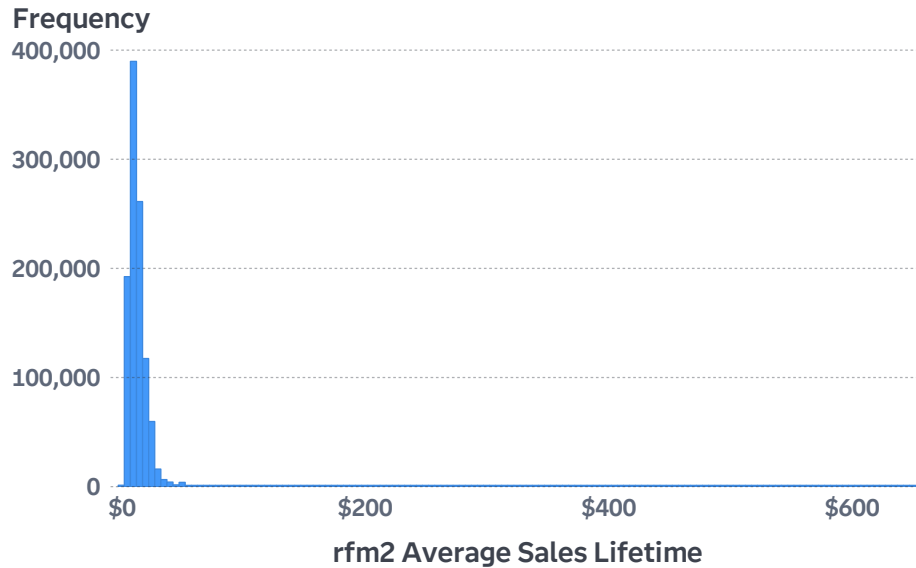
Response profile

Frequency Percent of tgt Binary New Product

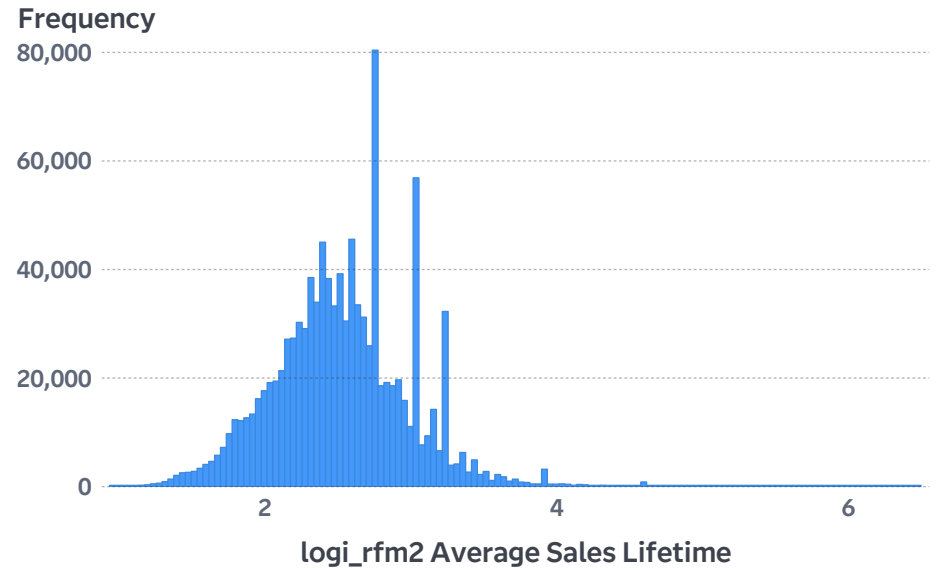


Distribution and Data Transformation

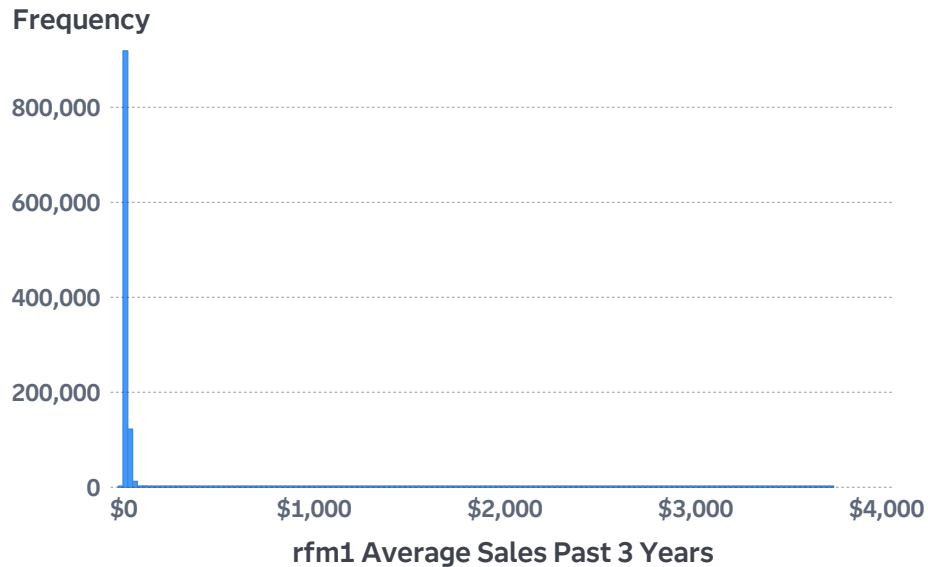
Frequency of rfm2 Average Sales Lifetime



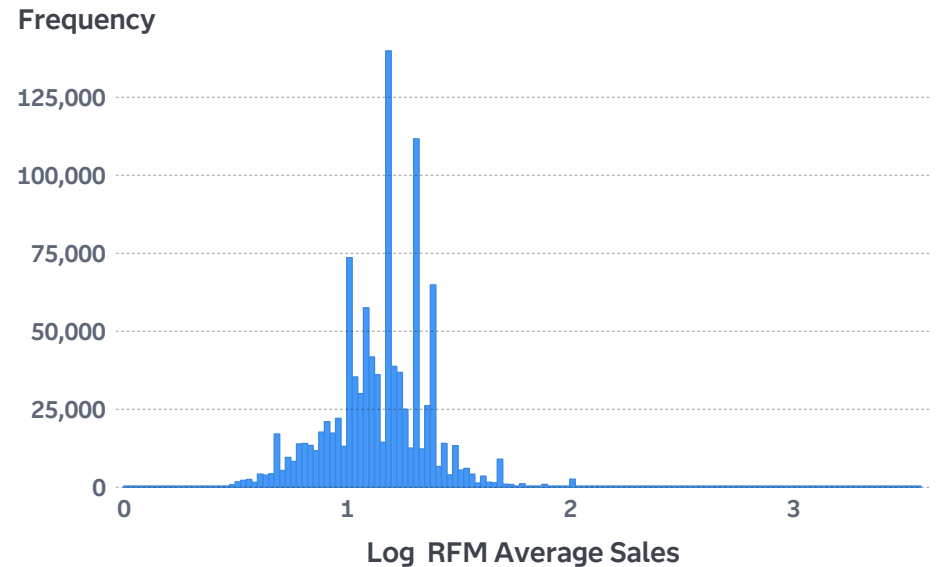
Frequency of logi_rfm2 Average Sales Lifetime



Frequency of rfm1 Average Sales Past 3 Years

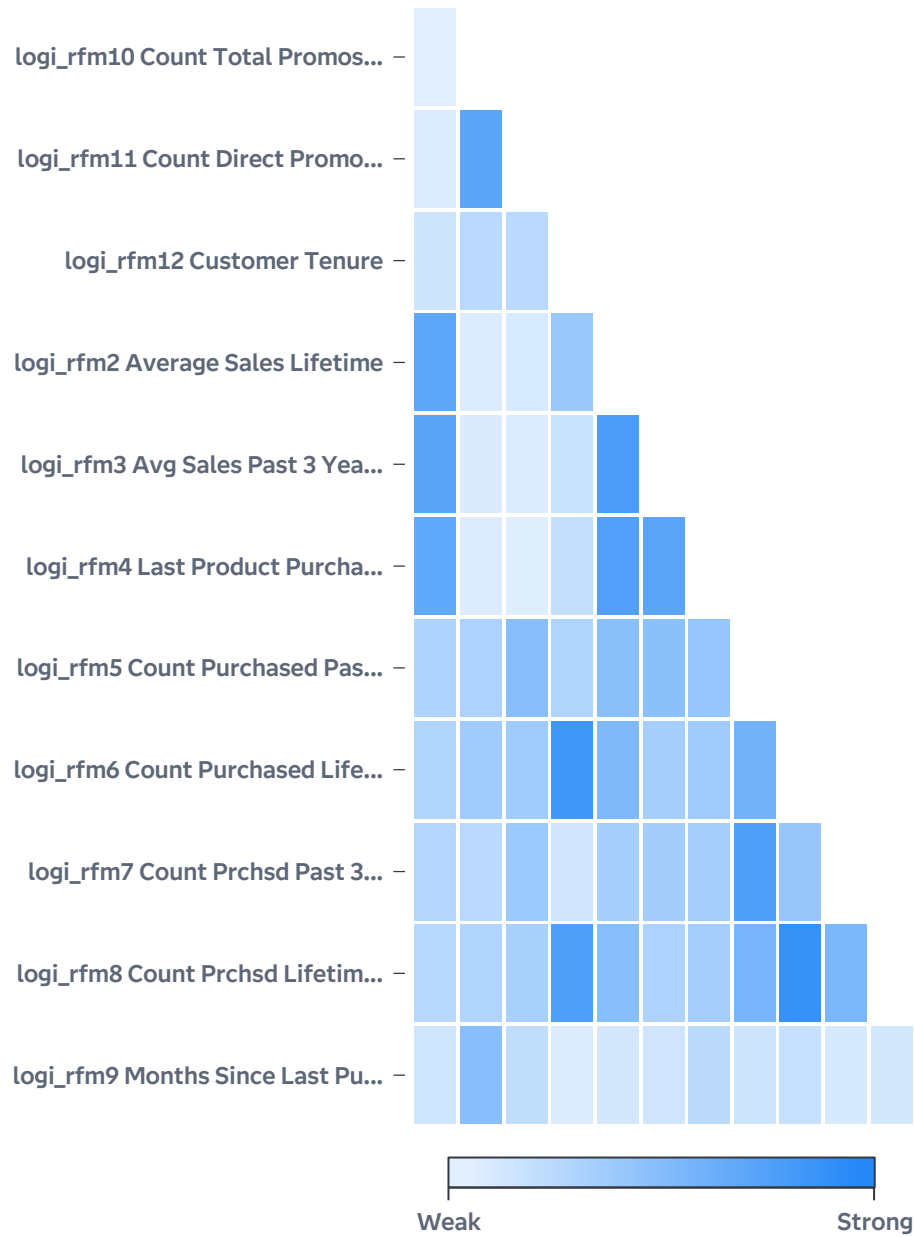


Frequency of Log RFM Average Sales

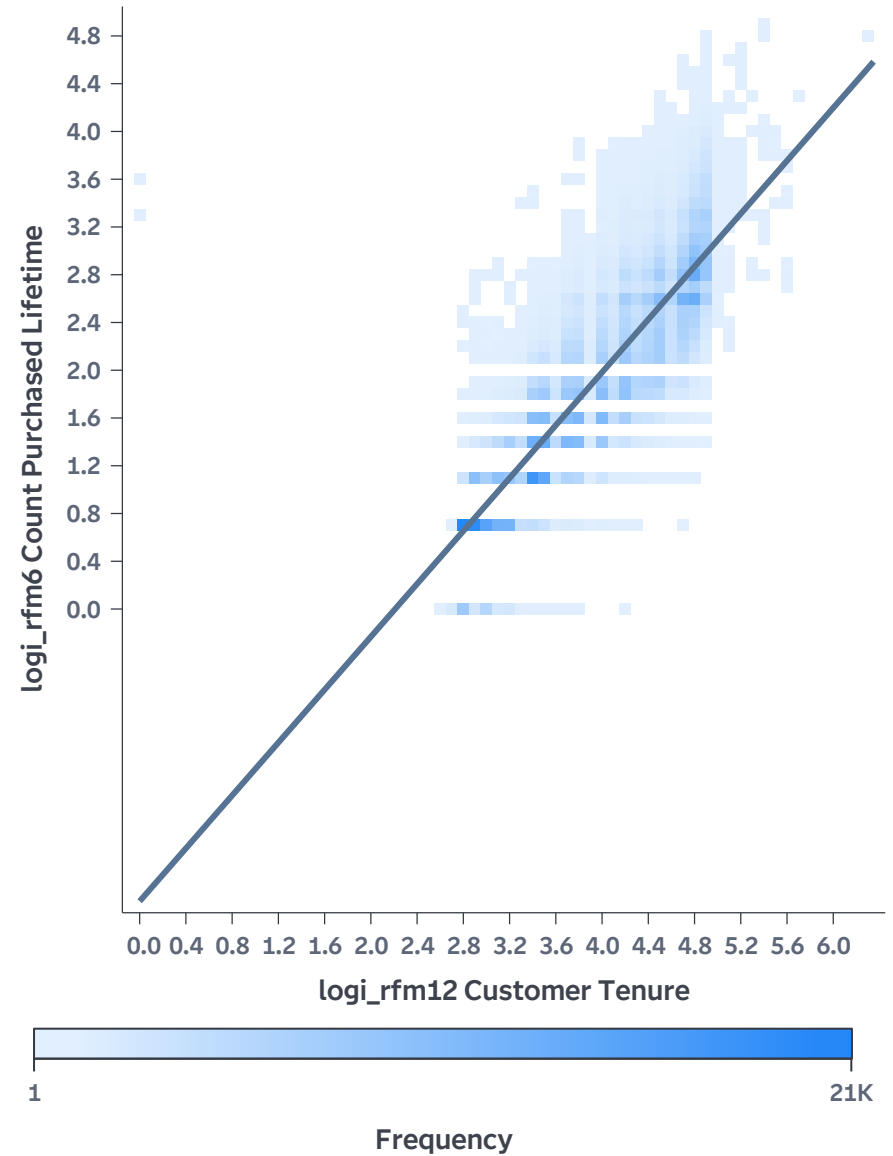


Correlation(redundant)

Correlation of Selected Measures



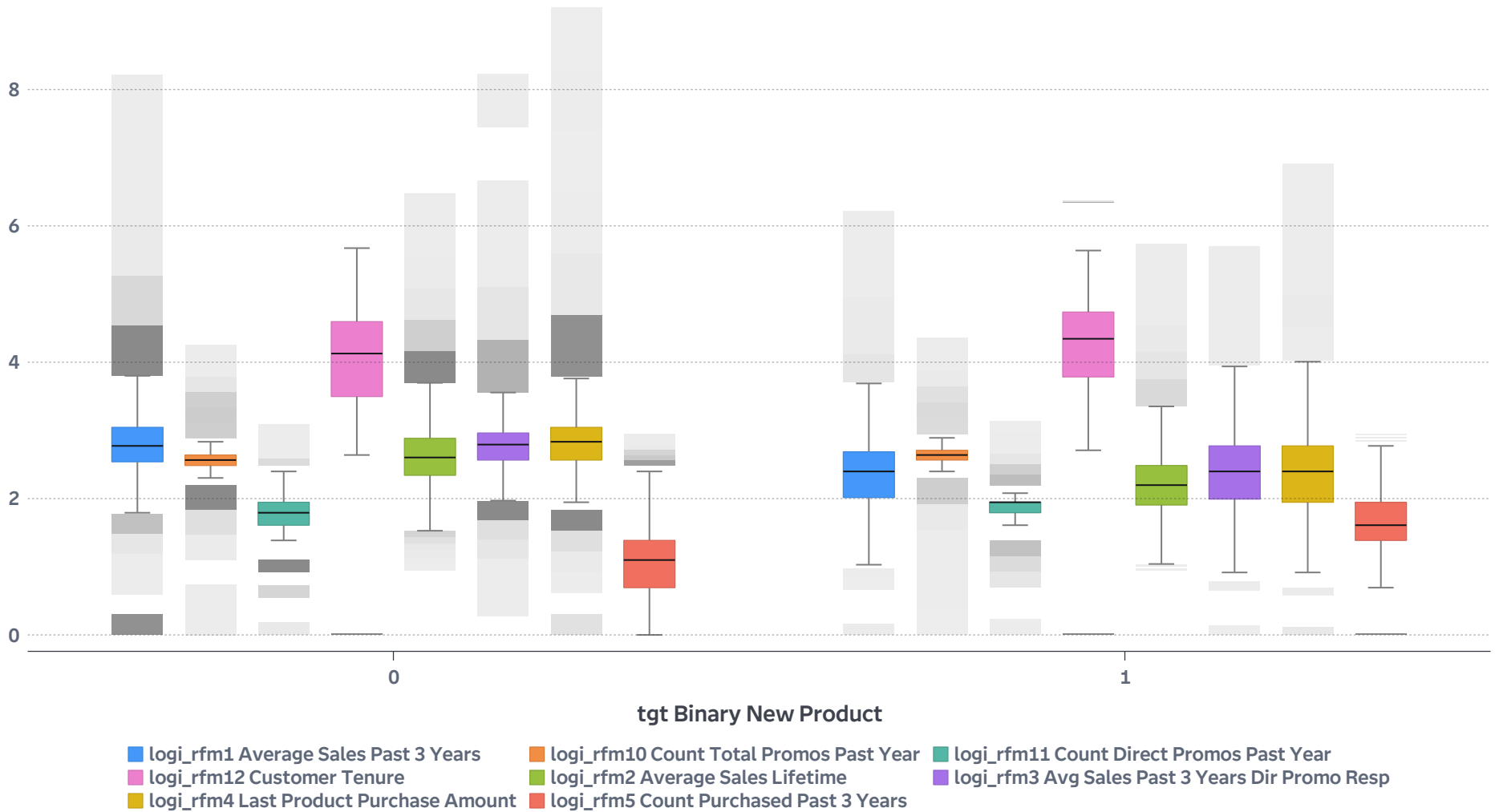
Frequency by logi_rfm12 Customer Tenure, logi_rfm6 Count Purchased Lifetime



Box Plot

logi_rfm1 Average Sales Past 3 Years, logi_rfm10 Count Total Promos Past Year, logi_rfm11 Count Direct Promos Past Year, logi_rfm12 Customer Tenure, logi_rfm2 Average Sales Lifetime, logi_rfm3 Avg Sales Past 3 Years Dir Promo Resp, logi_rfm4 Last Product Purchase Amount, logi_rfm5 Count Purchased Past 3 Years by tgt Binary New Product

Measures

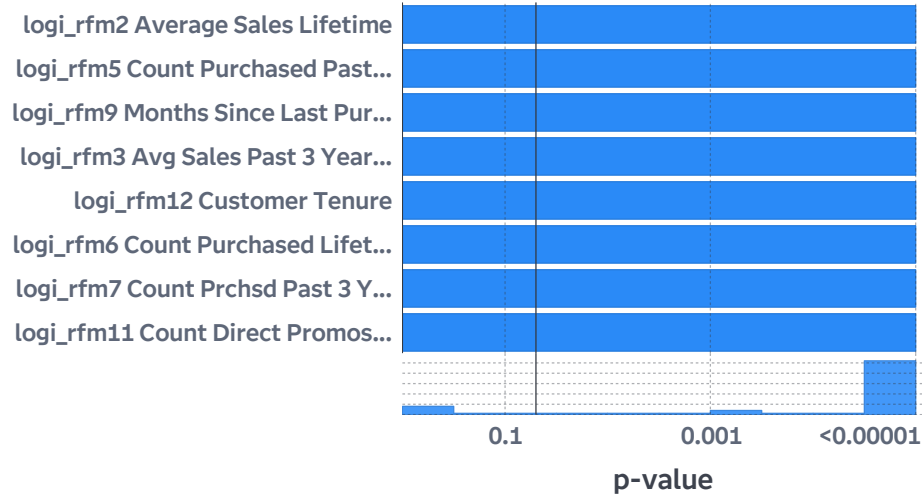


Logistic Regression Data

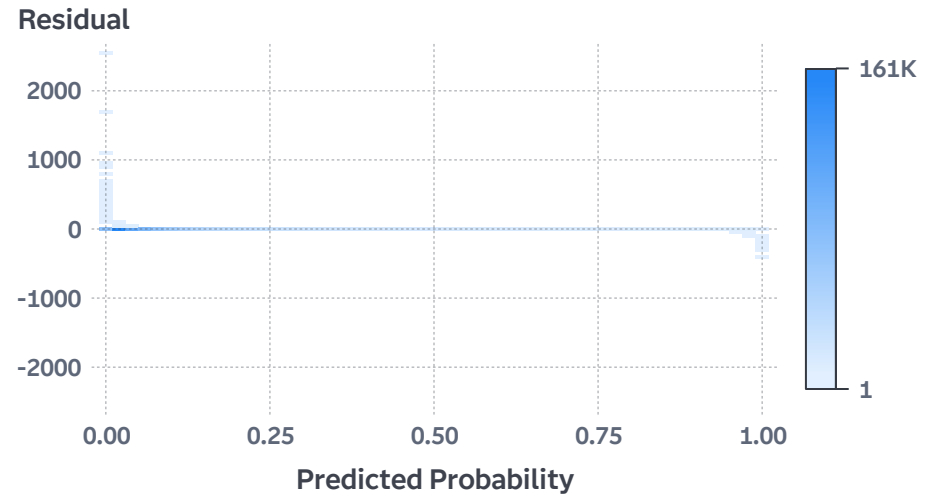
Logistic Regression of tgt Binary New Product

Event: 1 Fit: KS (Youden) 0.5844 Observations: 1.1M of 1.1M

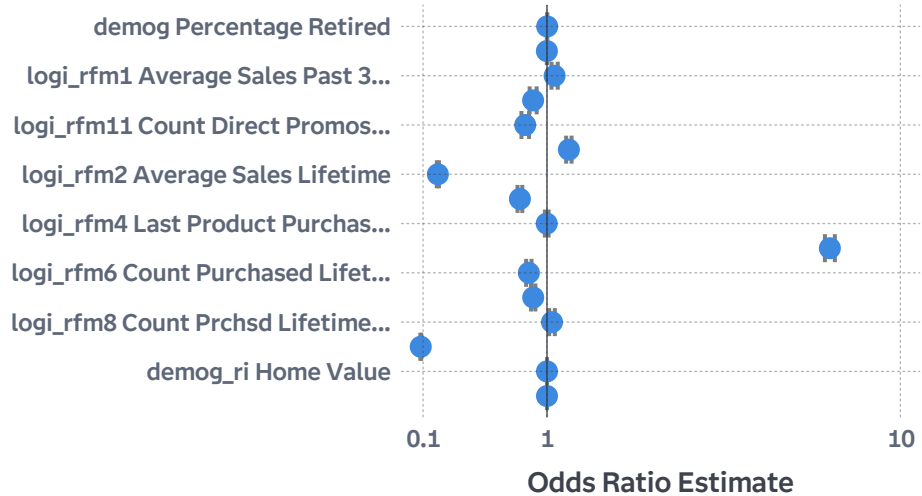
Fit Summary



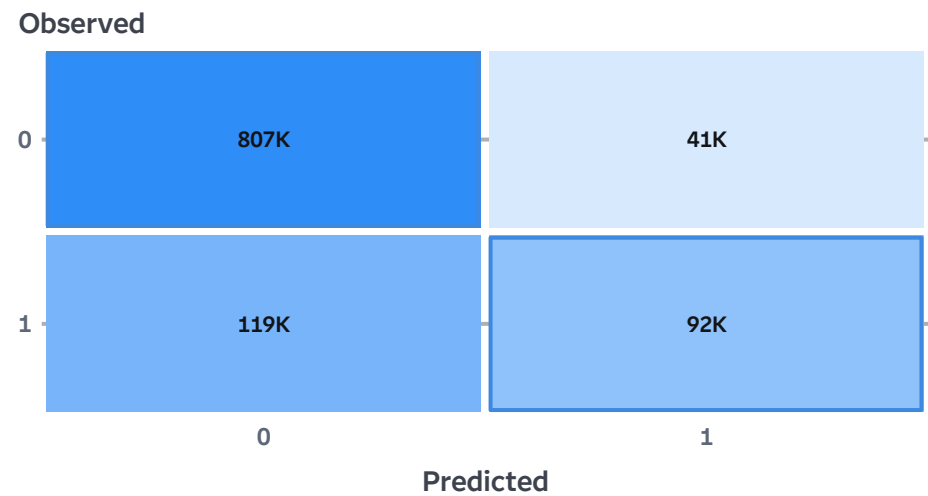
Residual Plot



Odds Ratio Plot



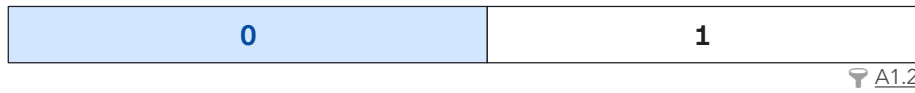
Confusion Matrix



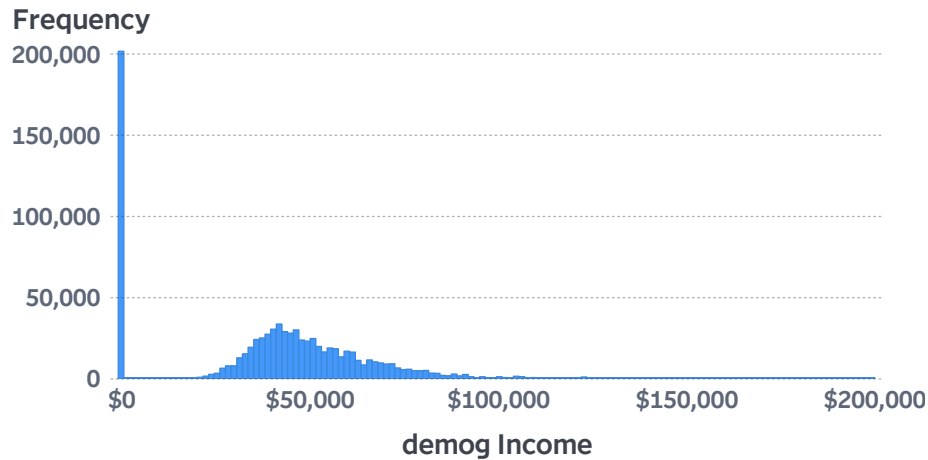
Model Prediction Scenarios

Predicted: tgt Binary New Product (1) ▲	Probability: tgt Binary New Product=1 (1)	Leaf ID (1)
0	536.28	32
0	418.61	46
0	26.63	56
0	1,121.42	52
0	17,780.49	38
0	6,254.64	37
0	9.26	59
0	2,785.12	43
0	10,410.30	41
0	352.55	40
0	35.95	39
0	4,892.87	49
0	17,841.52	31
0	116.25	47
0	329.96	33
0	19.37	16
0	6,923.13	44
0	1,053.12	57
0	35,854.71	34
0	11.38	55
0	113.74	54
0	179.79	53
0	159.47	35
0	3,521.28	51
0	2,199.97	50
0	929.51	58

Demographic and Purchase Distribution

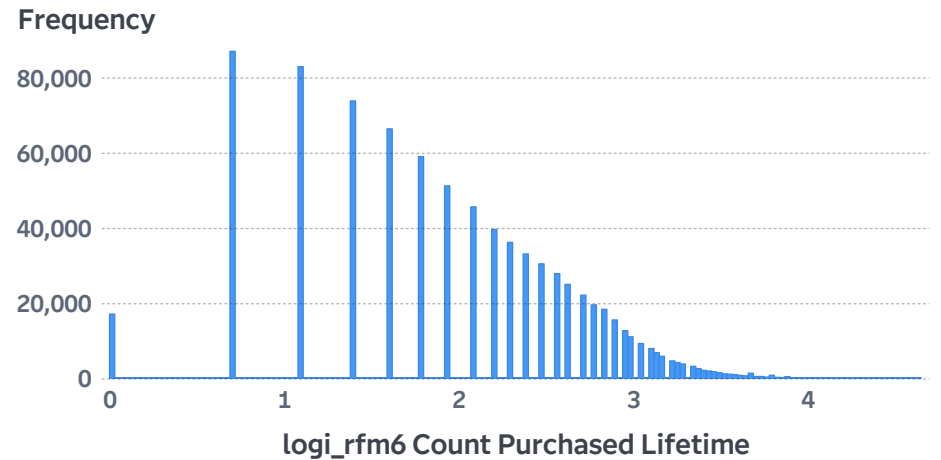


Frequency of demog Income



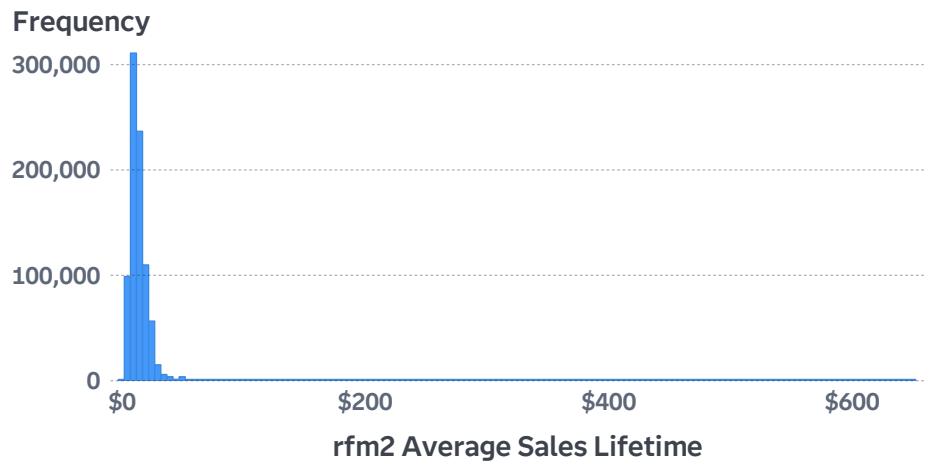
A1.3

Frequency of logi_rfm6 Count Purchased Lifetime



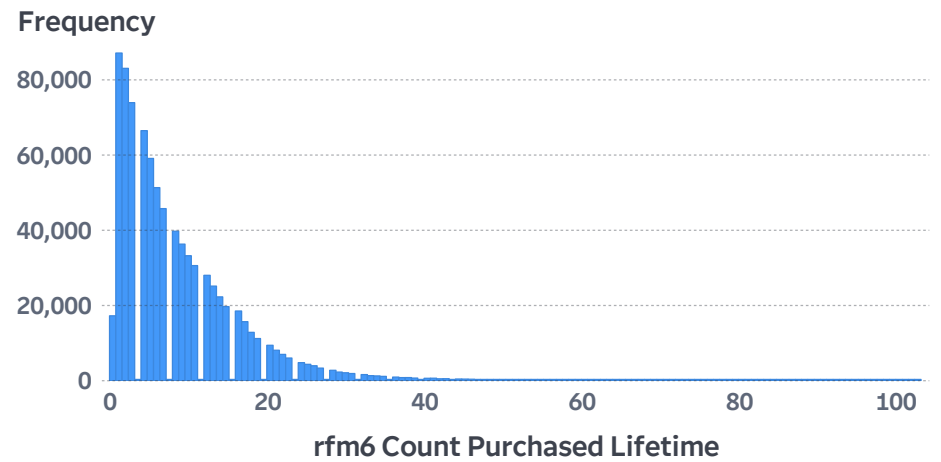
A1.4

Frequency of rfm2 Average Sales Lifetime



A1.5

Frequency of rfm6 Count Purchased Lifetime



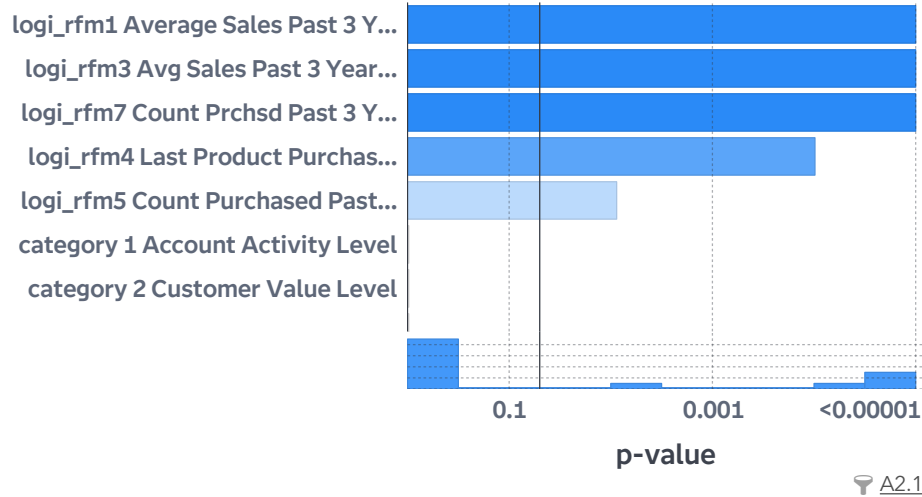
A1.6

Linear regression data

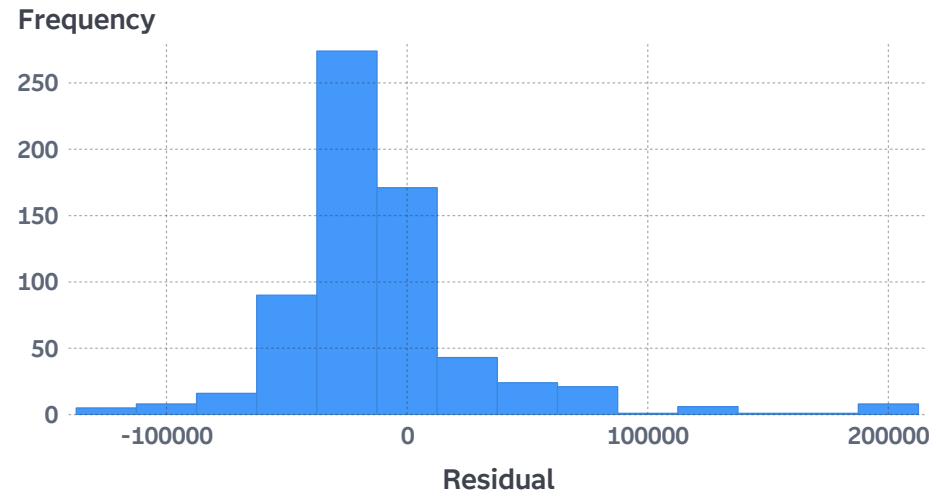
Linear Regression of tgt Interval New Sales

Fit: Validation ASE 2B Observations: 946 of 1.1M

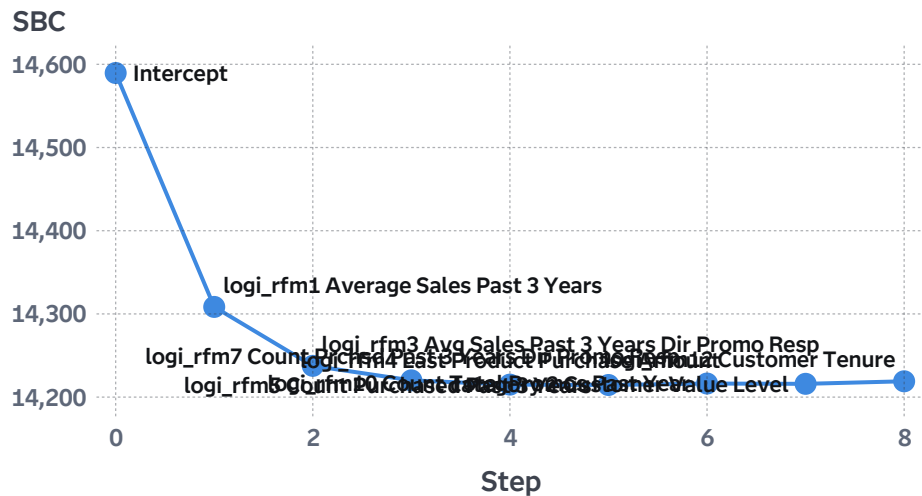
Fit Summary



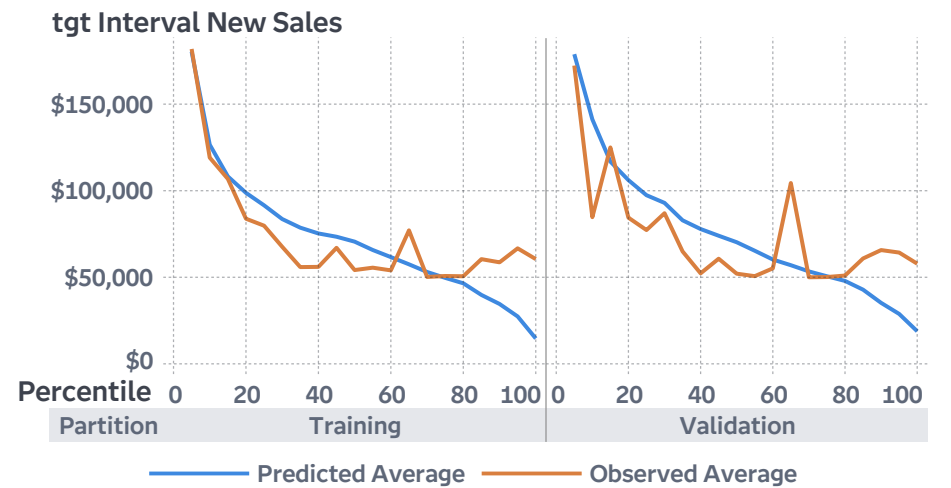
Residual Plot



Variable Selection Plot



Assessment

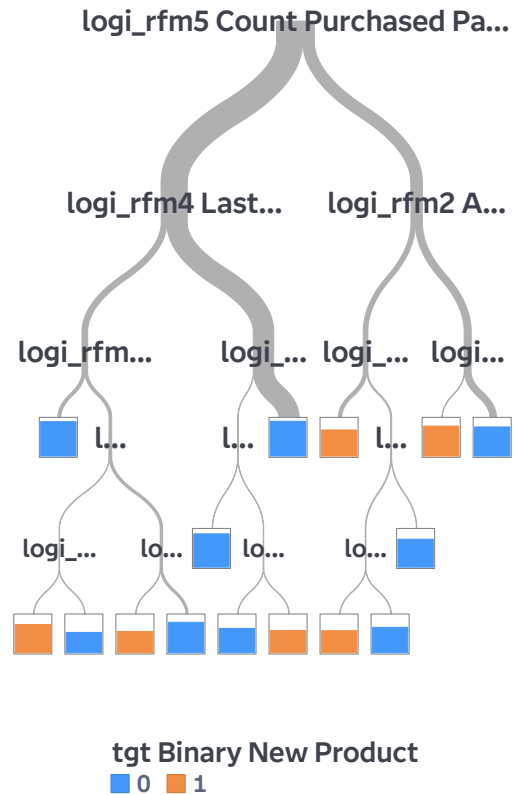


Classification Decision Tree

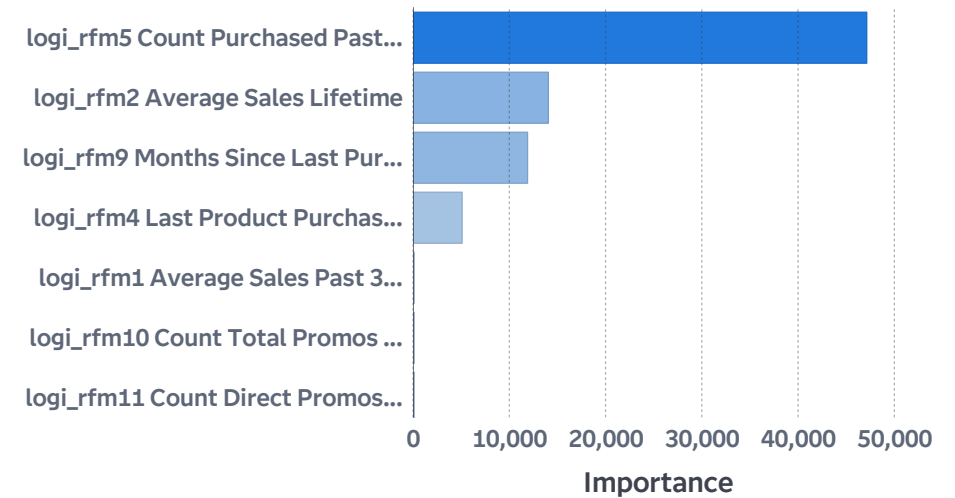
Decision Tree of tgt Binary New Product

Event: 1 Fit: KS (Youden) 0.4747 Observations: 1.1M of 1.1M

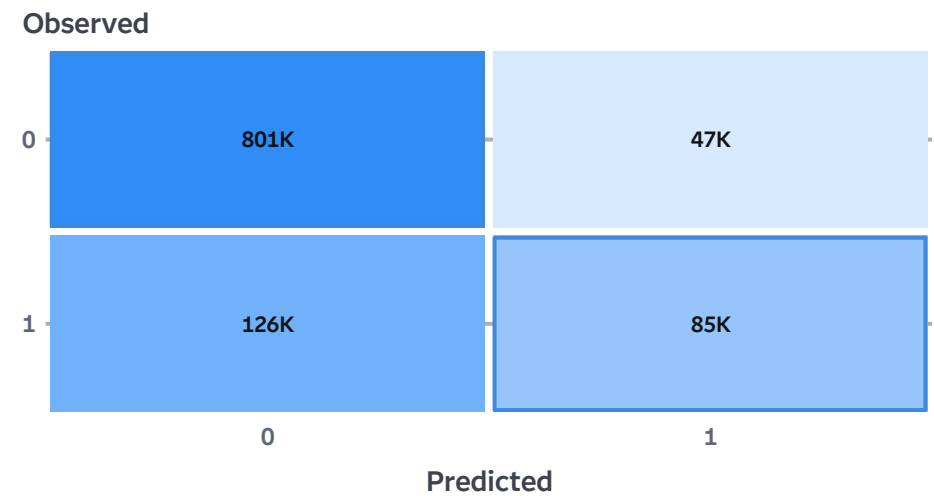
Tree



Variable Importance



Confusion Matrix



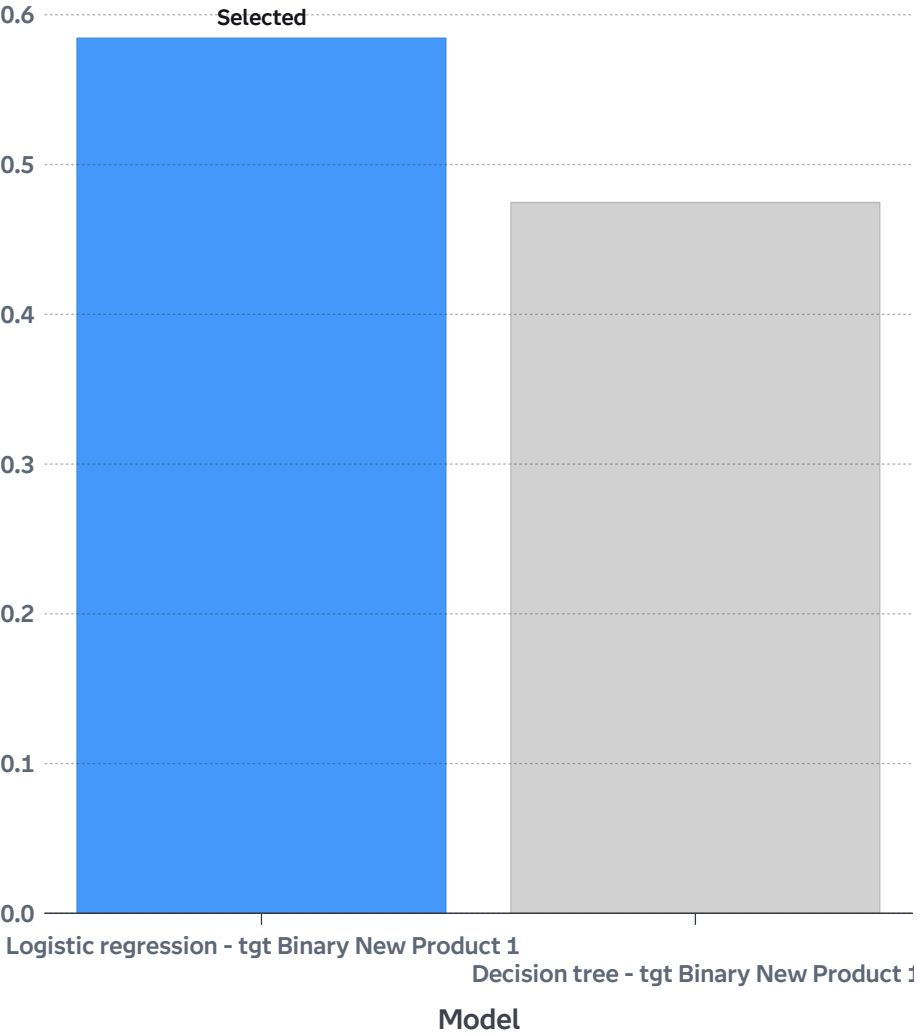
Model Comparison

Model Comparison of tgt Binary New Product

Event: 1

Fit Statistic

KS (Youden)



Confusion Matrix



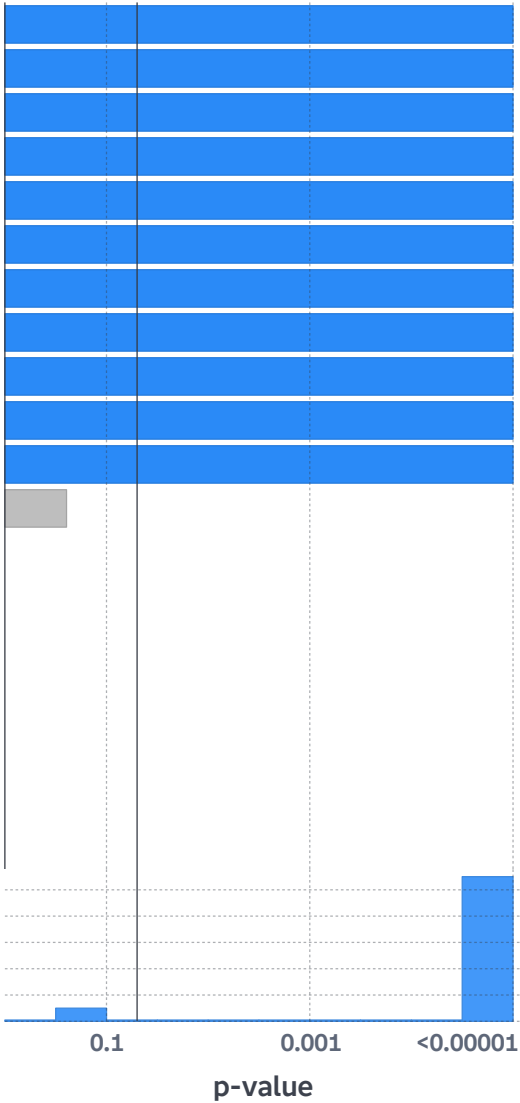
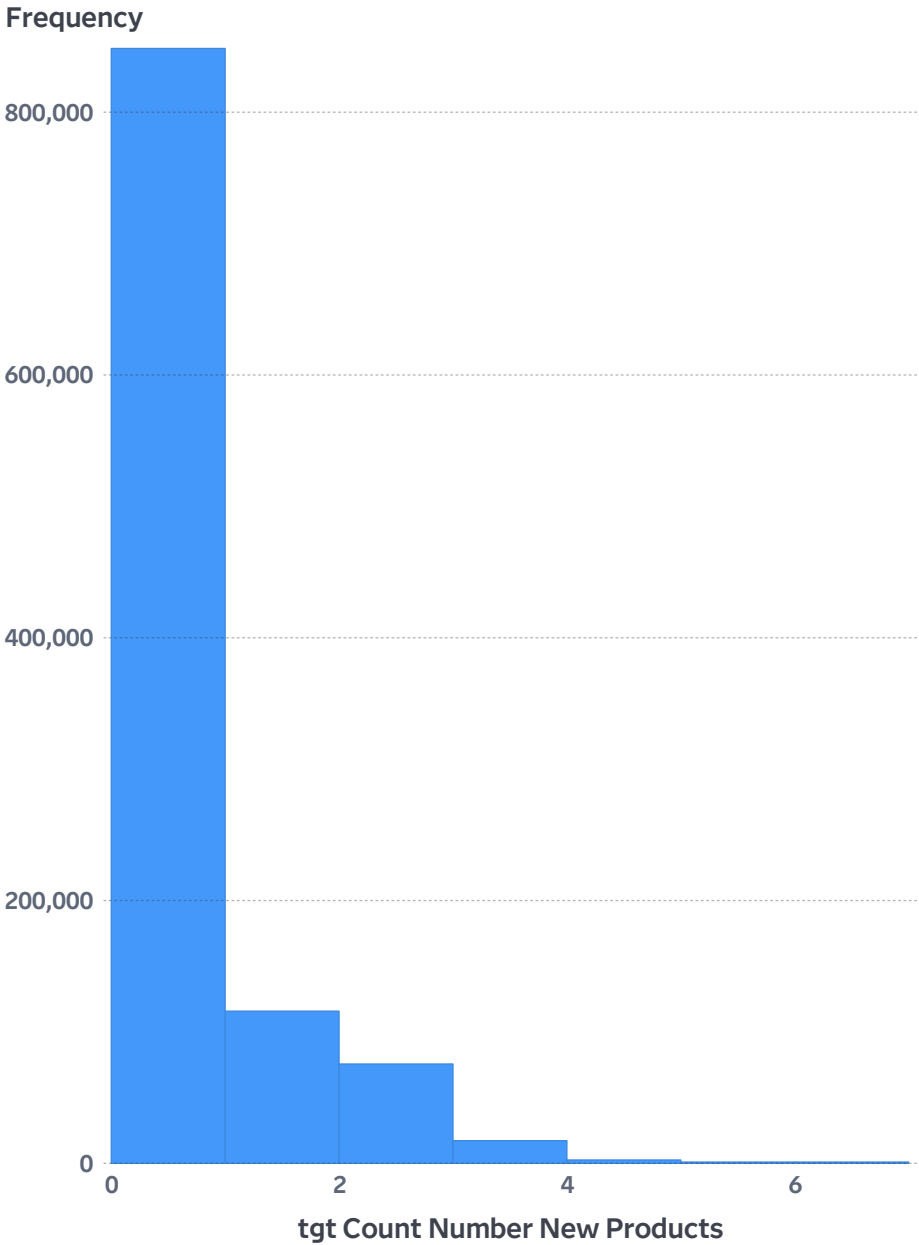
Generalized Linear Model of tgt Count Number New Prodi

Fit: ASE 0.35 Observations: 1.060037M of 1.060038M

Fit Summary

logi_rfm1 Average Sales Past 3 Y...	
logi_rfm11 Count Direct Promos...	
logi_rfm3 Avg Sales Past 3 Year...	
logi_rfm5 Count Purchased Past...	
logi_rfm7 Count Prchsd Past 3 Y...	
logi_rfm9 Months Since Last Pur...	
logi_rfm2 Average Sales Lifetime	
logi_rfm6 Count Purchased Lifet...	
logi_rfm4 Last Product Purchas...	
logi_rfm8 Count Prchsd Lifetime...	
logi_rfm12 Customer Tenure	
logi_rfm10 Count Total Promos ...	

Frequency of tgt Count Number New Products

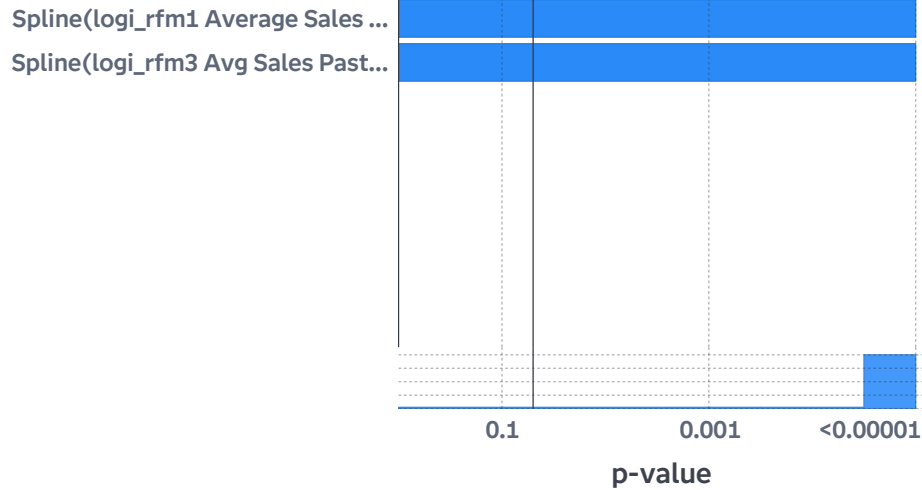


GAM

Generalized Additive Model of tgt Count Number New Products

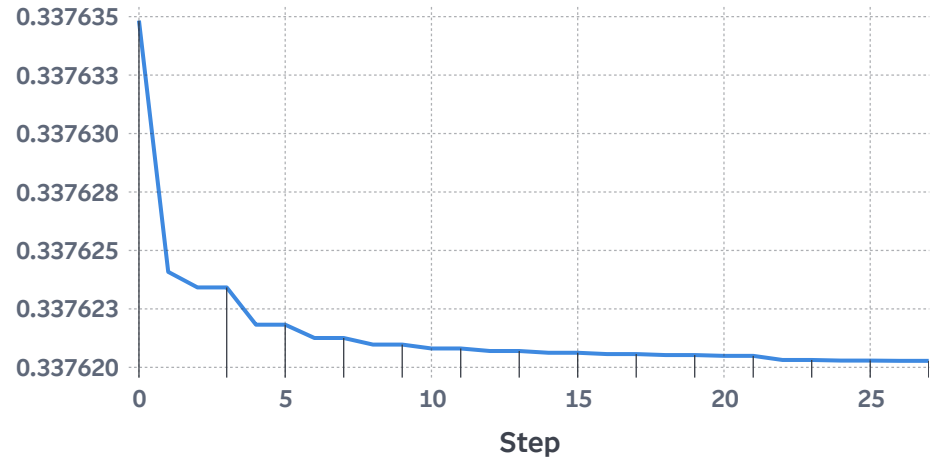
Fit: ASE 0.34 Observations: 1.060037M of 1.060038M

Fit Summary

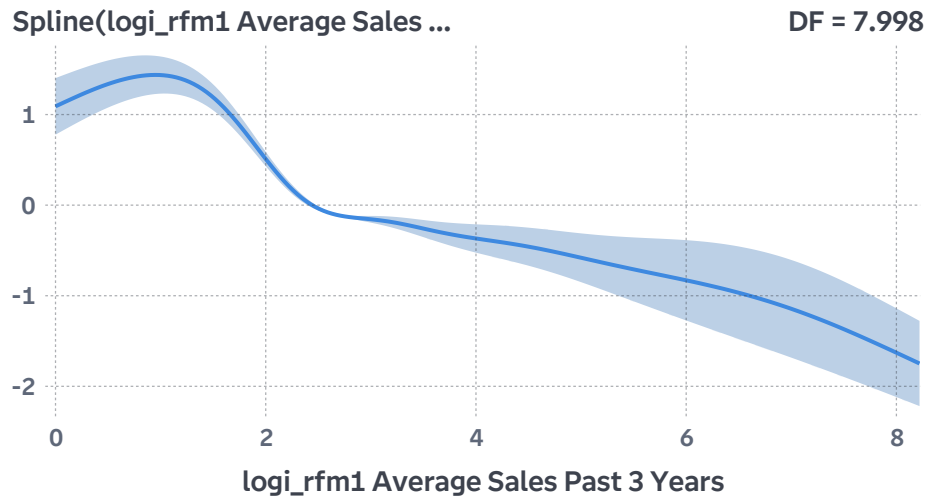


Iteration Plot

Objective Function

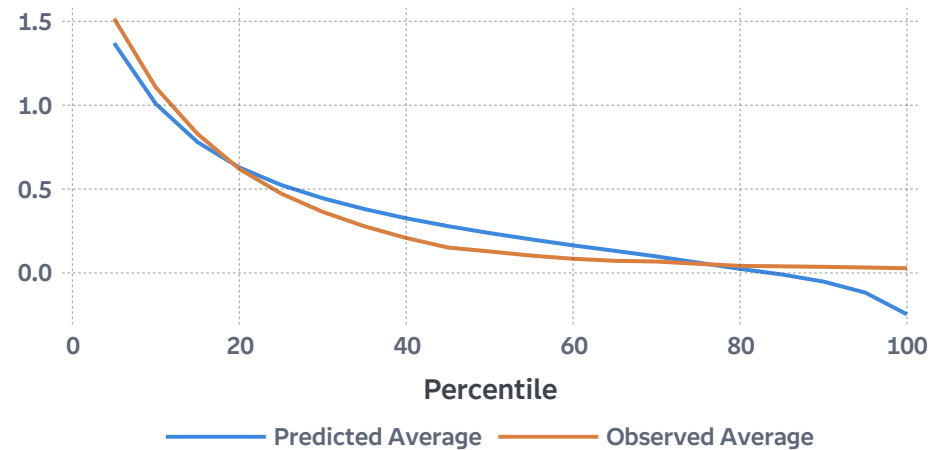


Spline Plot



Assessment

tgt Count Number New Products

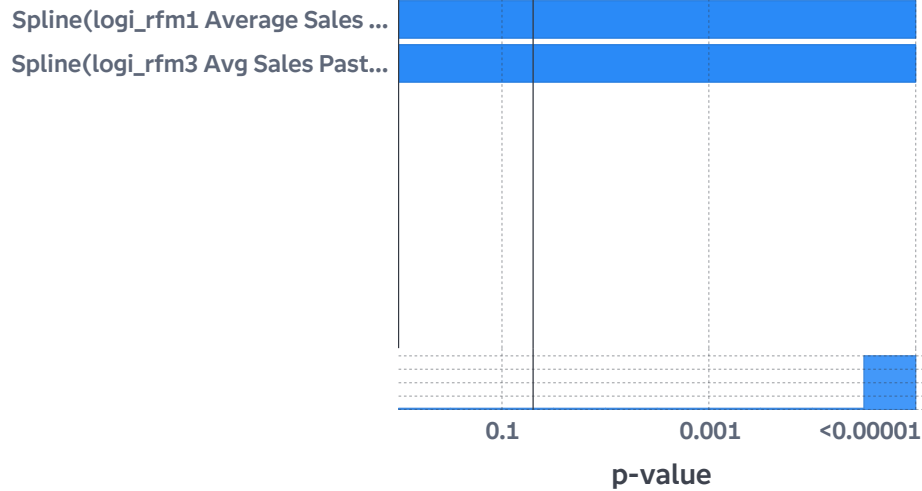


Non parametric

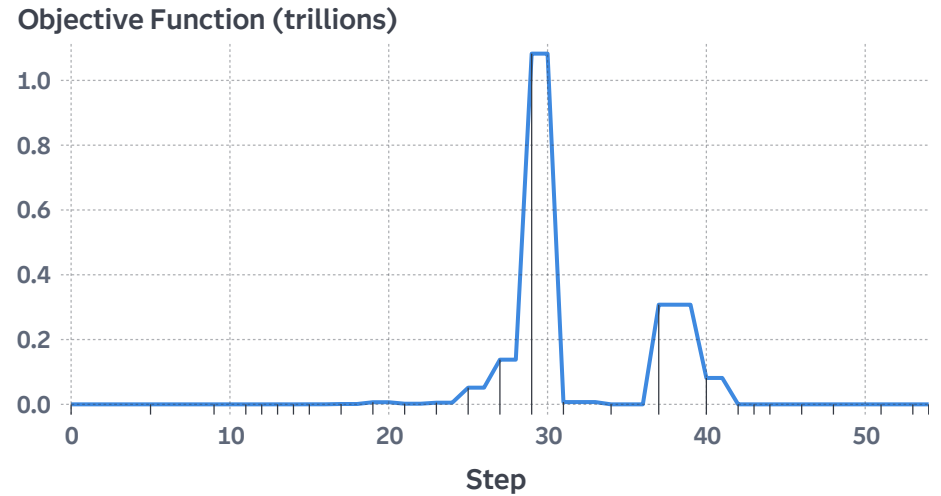
Nonparametric Logistic Regression of tgt Binary New Product

Event: 0 Fit: KS (Youden) 0.4159 Observations: 1.1M of 1.1M

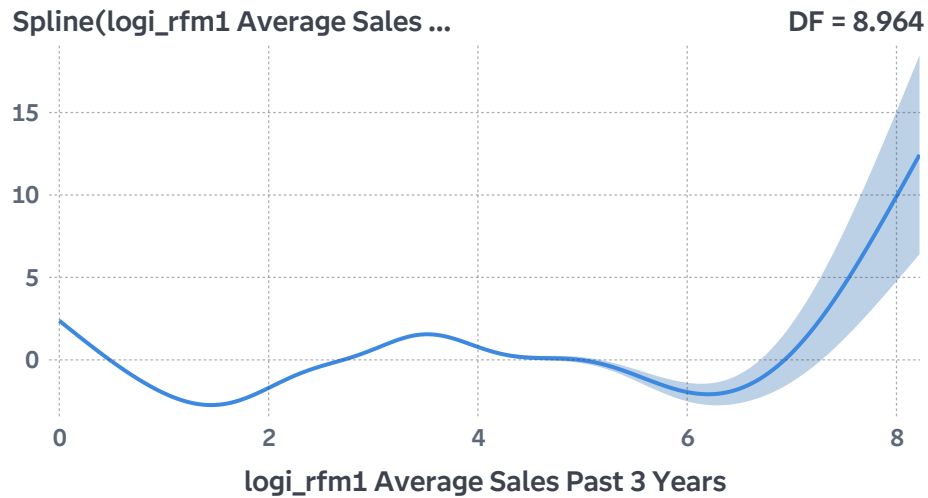
Fit Summary



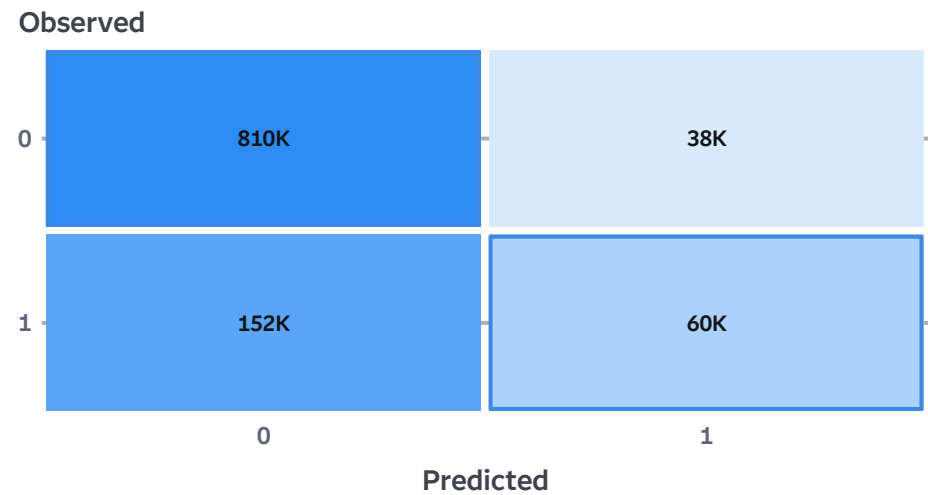
Iteration Plot



Spline Plot



Confusion Matrix

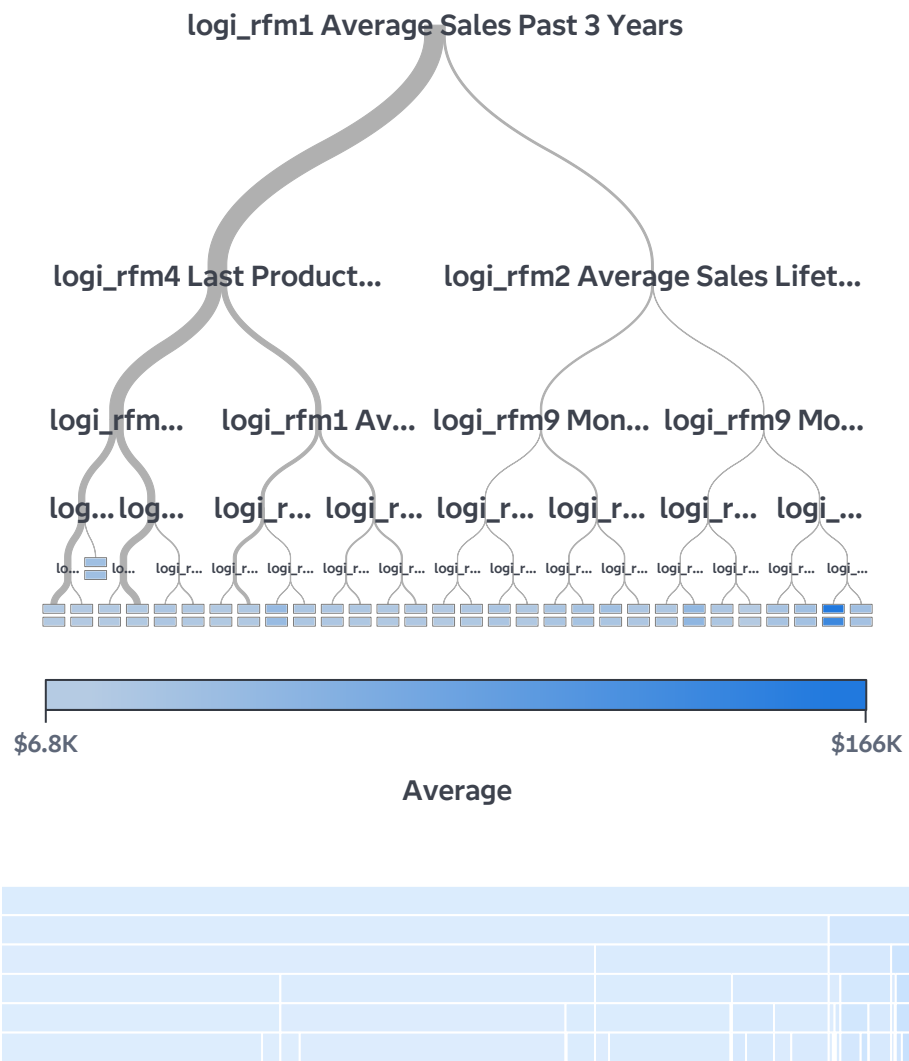


Decision TRee

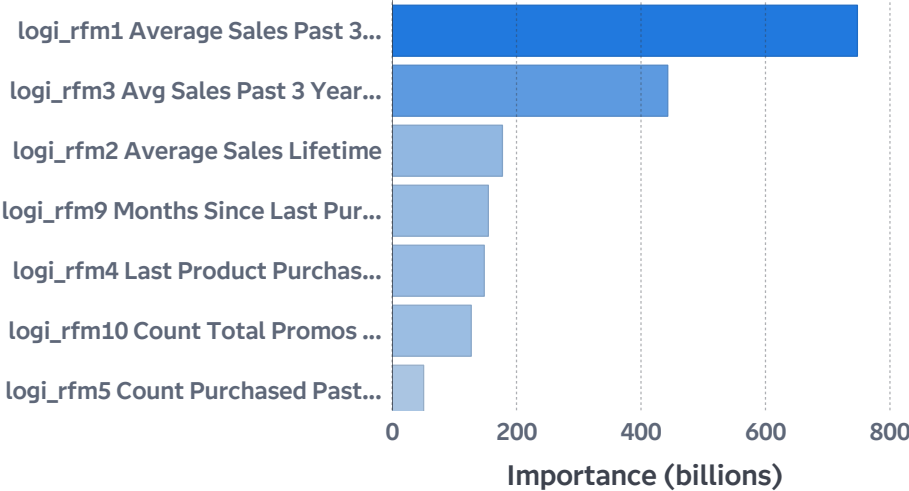
Decision Tree of tgt Interval New Sales

Fit: Validation ASE 60M Observations: 212K of 1.1M

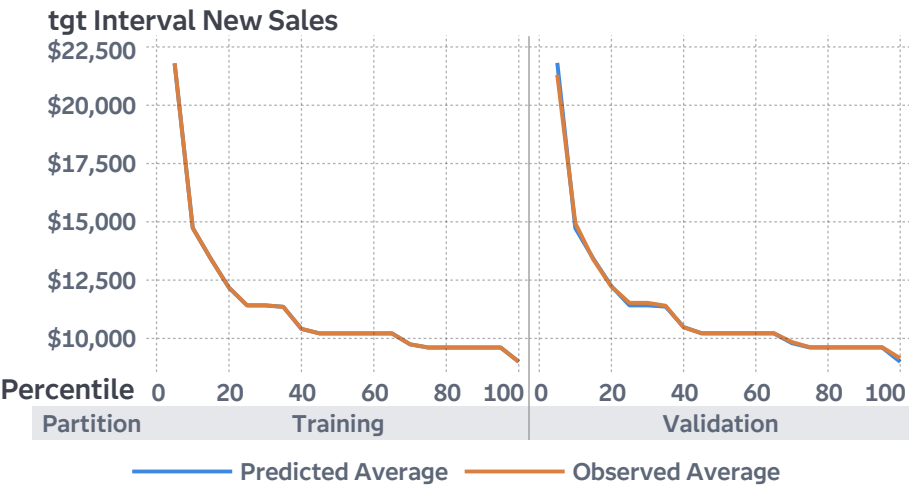
Tree



Variable Importance



Assessment



Regression Residuals and Leaf Analysis

Predicted: tgt Interval New Sales	tgt Interval New Sales ▲	Residual: tgt Interval New Sales	Leaf ID (1)
\$18,946.30	.	.	49
\$13,619.66	.	.	41
\$10,207.75	.	.	34
\$10,207.75	.	.	34
\$10,207.75	.	.	34
\$10,999.83	.	.	38
\$15,215.35	.	.	41
\$10,207.75	.	.	34
\$10,207.75	.	.	34
\$13,619.66	.	.	41
\$10,207.75	.	.	34
\$18,946.30	.	.	49
\$10,207.75	.	.	34
\$10,999.83	.	.	38
\$10,207.75	.	.	31
\$10,999.83	.	.	38
\$30,700.73	.	.	57
\$10,999.83	.	.	38
\$11,236.13	.	.	36
\$10,207.75	.	.	31
\$13,619.66	.	.	41
\$10,207.75	.	.	34
\$12,088.17	.	.	37
\$10,999.83	.	.	38
\$10,207.75	.	.	34
\$15,215.35	.	.	41

▲ A3.1

Appendix

A1.1 Bank Analysis

Parameters: Prediction cutoff for Predicted: tgt Binary New Product (1) = 0.50

A1.2 Button bar - tgt Binary New Product 1

Filters: $0.00 \leq \text{Probability: tgt Binary New Product}=1 (1) \leq 1.00$

A1.3 Frequency of demog Income

Filters: tgt Binary New Product = 0
 $0.00 \leq \text{Probability: tgt Binary New Product}=1 (1) \leq 1.00$

A1.4 Frequency of logi_rfm6 Count Purchased Lifetime

Filters: tgt Binary New Product = 0
 $0.00 \leq \text{Probability: tgt Binary New Product}=1 (1) \leq 1.00$

A1.5 Frequency of rfm2 Average Sales Lifetime

Filters: tgt Binary New Product = 0
 $0.00 \leq \text{Probability: tgt Binary New Product}=1 (1) \leq 1.00$

A1.6 Frequency of rfm6 Count Purchased Lifetime

Filters: tgt Binary New Product = 0
 $0.00 \leq \text{Probability: tgt Binary New Product}=1 (1) \leq 1.00$

A2.1 Fit Summary

Filters: (\$50,000.00 $\leq$ tgt Interval New Sales $\leq$ \$500,000.00) OR tgt Interval New Sales MISSING

A3.1 List table - Predicted: tgt Interval New Sales 1

Warnings: Only 40,000 rows of the data appear.